ANY FIELD POINT OF A SPACE ADMITTING A FLAT FIELD COVER IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3060-3075, immediately following the statement at 3052-3058.

Editorial note. One cross-chapter pointer (Morphisms of Spaces) is recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Flatness permanence under composition. Morphisms of Spaces, Lemma spaces-morphisms-lemma-flat-permanence, cited at proof lines 3070-3072. Composition T -> Spec(k') -> Z with both flat gives Spec(k') -> Z flat.
- Fibred products over a field are flat. field-case flatness invoked at proof lines 3066-3067 (T -> Spec(k) is flat as k is a field). Every morphism to the spectrum of a field is flat.

Source reference. Original reference 05VY: focused-seeds/geometry/S9-decent-flat-cover-by-field/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let Z be an algebraic space over S. Let k be a field and let $\mathrm{Spec}(k) \to Z$ be surjective and flat. Then any morphism $\mathrm{Spec}(k') \to Z$ where k' is a field is surjective and flat.*

Proof. Consider the fibre square

$$\begin{array}{ccc} T & \longrightarrow & \mathrm{Spec}(k) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k') & \longrightarrow & Z \end{array}$$

Note that $T \to \mathrm{Spec}(k')$ is flat and surjective hence T is not empty. On the other hand $T \to \mathrm{Spec}(k)$ is flat as k is a field. Hence $T \to Z$ is flat and surjective. It follows from Morphisms of Spaces, Lemma 05VY that $\mathrm{Spec}(k') \to Z$ is flat. It is surjective as by assumption $|Z|$ is a singleton. □

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